

Exercise 4: Launching Web Applications using App Engine Tools

Aim:

To launch and run a web application locally using Google App Engine tools (Flask) and verify it in a browser.

Procedure:

Step 1: Upgrade Flask and Werkzeug

```
pip install --upgrade flask werkzeug
```

This upgrades Flask and its dependencies (Werkzeug, blinker, etc.) in the virtual environment. Step

2: Verify Installed Package Versions

```
pip show flask pip  
show werkzeug
```

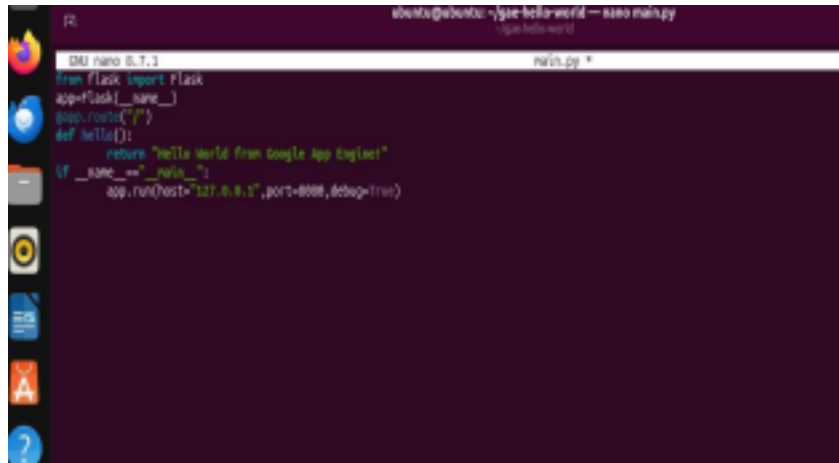
```
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show flask  
Name: Flask  
Version: 2.0.1  
Summary: A simple framework for building complex web applications.  
Home-page: https://palletsprojects.com/p/Flask  
Author: Armin Ronacher  
Author-email: armin.ronacher@active4.com  
License: BSD-3-Clause  
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages  
Requires: click, itsdangerous, Jinja2, Werkzeug  
Required-by:  
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show werkzeug  
Name: Werkzeug  
Version: 3.1.1  
Summary: The comprehensive WSGI web application library.  
Home-page:  
Author:  
Author-email:  
License-Expression: BSD-3-Clause  
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages  
Requires: MarkupSafe  
Required-by: Flask  
(venv) ubuntu@ubuntu:~/gae-hello-world$ python main.py
```

Step 3: Review Application Code (main.py)

main.py defines a simple Flask route that returns a Hello World message:

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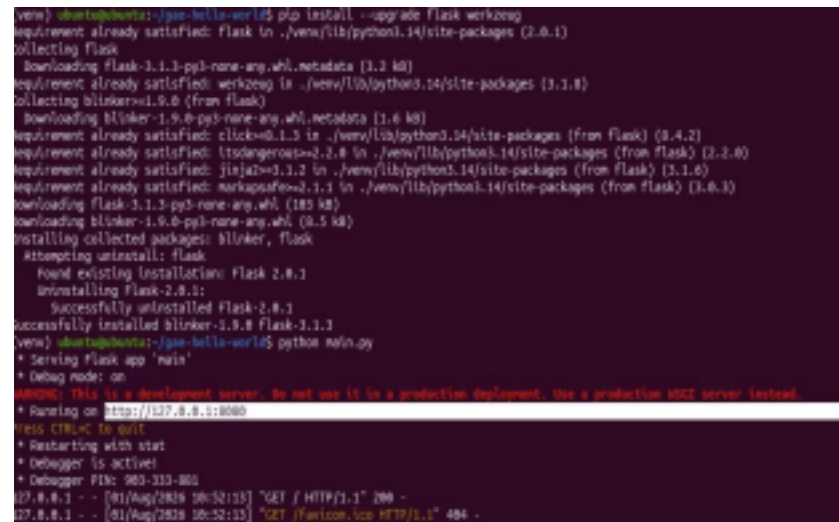


```
main.py *
from flask import Flask
app=Flask(__name__)
app.run(host='127.0.0.1',port=8080,debug=True)
def hello():
    return "Hello world from Google App Engine!"
if __name__=="__main__":
    app.run(host='127.0.0.1',port=8080,debug=True)
```

Step 4: Run the Application

python main.py

Flask starts the development server on <http://127.0.0.1:8080> with debug mode enabled.

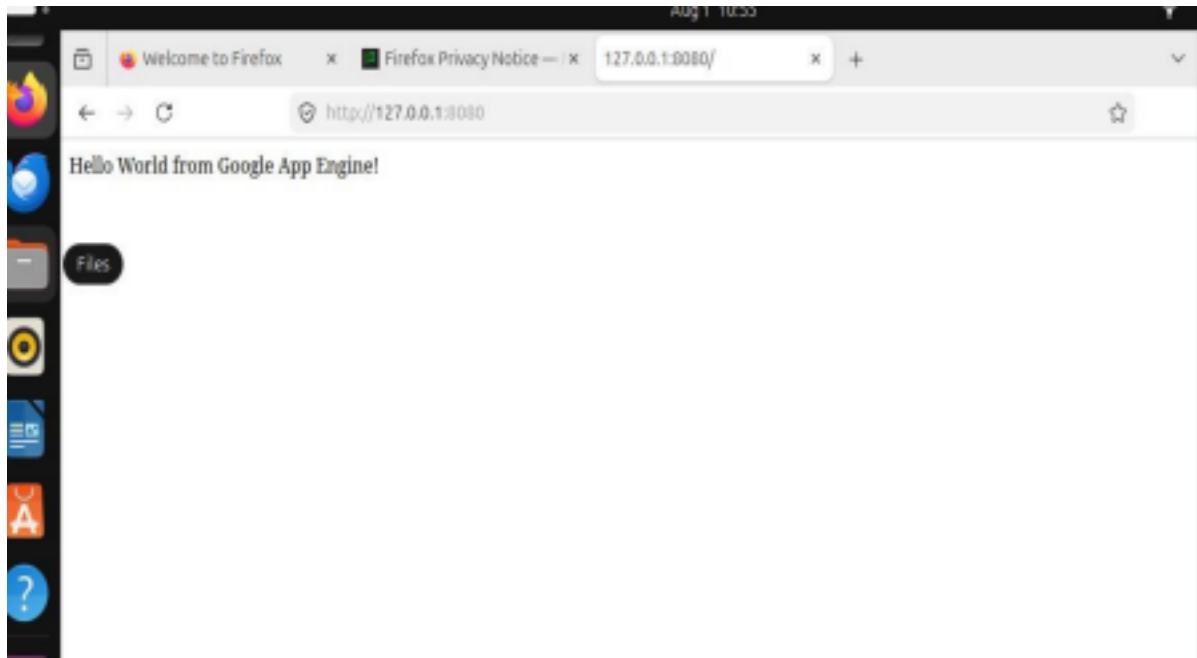


```
(venv) ubuntu@ubuntu:~/jpa-hello-world$ pip install --upgrade flask werkzeug
Requirement already satisfied: flask in ./venv/lib/python3.14/site-packages (2.0.1)
Collecting flask
  Downloading flask-1.1.1-py3-none-any.whl.metadata [3.2 kB]
Requirement already satisfied: werkzeug in ./venv/lib/python3.14/site-packages (3.1.0)
Collecting blinker<1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata [1.6 kB]
Requirement already satisfied: click<8.1.3 in ./venv/lib/python3.14/site-packages (from flask) (8.4.2)
Requirement already satisfied: itsdangerous<2.2.0 in ./venv/lib/python3.14/site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2<3.1.2 in ./venv/lib/python3.14/site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe<2.1.1 in ./venv/lib/python3.14/site-packages (from flask) (2.0.1)
Downloading flask-1.1.1-py3-none-any.whl (183 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Installing collected packages: blinker, flask
  Attempting uninstall: flask
    Found existing installation: Flask 2.0.1
    Uninstalling Flask-2.0.1:
      Successfully uninstalled Flask-2.0.1
Successfully installed blinker-1.9.0 flask-1.1.1
(venv) ubuntu@ubuntu:~/jpa-hello-world$ python main.py
* Serving Flask app 'main'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8080
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 903-333-860
127.0.0.1 - - [01/Aug/2025 10:52:15] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Aug/2025 10:52:15] "GET /favicon.ico HTTP/1.1" 404 -
```

Output:

The application was accessed from the browser at 127.0.0.1:8080, confirming successful deployment:

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Result:

Thus, the Flask-based web application was successfully launched and tested locally using Google App Engine tools.